

Kubenet network plugin (retired in 2028)

- Network plugin: software component that provides networking functionality to the cluster
- Default network plugin in AKS; also known as basic network plugin
- IP address assignment:
 - Nodes: from AKS subnet
 - Pods: from pod CIDR, which is a logical address space
- The pod CIDR is broken into smaller pieces and assigned to nodes
- Standard routing table and Address Resolution Protocol (ARP) facilitate pod-to-pod communication within the same node
- An Azure Route Table is required to facilitate the communication between pods on different nodes
- Pod's IP is NAT'd to node's IP for pods to reach resources on the Azure VNET or outside it

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10.224.0.0/12

AKS VNET



10.224.0.0/16

AKS Subnet

Node 0: 10.224.0.4



10.244.0.0/24

Pod A



10.244.0.15

Internal
Table

Pod B



10.244.0.16

Address
prefix

Next hop

10.244.0.0/24

10.224.0.4

10.244.1.0/24

10.224.0.5

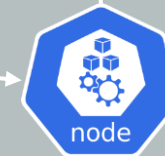
AKS Route Table

Pod CIDR
10.244.0.0/16



10.224.1.2

NATing
Node 1: 10.224.0.5



NATing

10.244.1.0/24

Pod C



10.244.1.27

Another VNET
/ On prem



10.1.2.3

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- **Why would you choose kubenet?**
 - Requires few IP addresses
 - Good option if the pods' communication is predominant within the cluster
- **What disadvantages apply to kubenet?**
 - There are additional networking hops that add a minor latency
 - You will need a route table, which can have a maximum of 400 routes, meaning you can have up to 400 nodes in a cluster
 - You cannot have multiple kubenet clusters in the same subnet
 - It doesn't support Windows nodes, Azure Network Policy, Virtual Nodes